DermTrace: Evidence-Grounded Agents for Vision-Language Diagnosis in Dermatology

A data curation report and a research proposal

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Technical report, 2026. Builds on remote work with a CMU–Yale team in 2025. Not peer reviewed.

Abstract

Vision-language models (VLMs) can now write fluent dermatology diagnoses from a photograph and a short history. Whether those diagnoses rest on the lesion in the image is a separate question, and current evaluations rarely answer it. This report has two parts. The first describes the data work I did for a CMU–Yale evaluation of LLM and VLM diagnosis: turning PubMed Central (PMC) dermatology case reports into image-text case records that can be traced back to their source and that do not leak the diagnosis through captions. The second proposes *DermTrace*, a framework that connects a VLM, an LLM and a small set of tool-using agents so that every diagnostic claim is tied either to a region of the image or to a cited reference case, and is checked before it is shown. Its three ideas are: describe findings before naming a diagnosis, test each visual finding with a counterfactual that hides the region it points to, and retrieve reference cases by shared diagnostic criteria rather than by image similarity alone. The report closes with research questions, an evaluation plan, and the risks I expect. No model results are reported; the team's evaluation results are not mine to report.

1 Motivation

Dermatology is a natural test bed for medical VLMs: the key evidence is visible, and the field has a precise vocabulary for describing it (lesion type, color, border, distribution, configuration). Benchmarks built from published case reports have made rigorous evaluation possible. MedR-Bench restructures case reports from the PMC Open Access Subset into 1,453 patient cases and scores diagnostic reasoning against reference steps [1]; DermCase builds a long-context benchmark of rare skin disease from licensed journal case reports [2]; DermBench pairs a benchmark of diagnostic narratives with a trained multimodal evaluator [3].

Three findings from recent work motivate this proposal. First, when the correct medical image is replaced with a blank placeholder, several frontier VLMs lose only a few points of accuracy and still describe findings they never saw [4]. Second, medical VLMs often attend to text cues rather than to the region that carries the evidence [5], and training for accuracy alone can raise scores while weakening the link between the answer and the image [6]. Third, multi-agent LLM systems can improve medical decision-making by assigning distinct roles and adding a review step [7, 8], but they have mostly been studied on text questions, where there is no image to be grounded in. A recent position paper argues for the same direction in physiological signals: moving from prediction toward verifiable reasoning with foundation models and agentic LLMs [9].

Put together: dermatology needs systems that use agents not only to reason more, but to *check* that the reasoning is anchored in what is visible and in citable prior cases.

2 What I built: a leakage-aware, traceable case corpus

2.1 Scope

The CMU–Yale project evaluated LLM and VLM diagnoses on dermatology case reports; the team's pipeline scores factual accuracy, step-by-step reasoning and evidence traceability. My part was the

data layer: crawling and curating PMC dermatology case reports, aligning each clinical image with its caption and source article, tracking licenses, masking captions that reveal the diagnosis, and integrating the corpus into the team's pipeline. I did not design the team's evaluator.

2.2 Pipeline

Articles are located through PubMed and resolved to PMC identifiers (PMCIDs). For each article, the pipeline retrieves and parses the full text, figures, captions and license, and stores them in structured form. Collection is modular, with one collector per journal, so that journal-specific layouts stay contained. A SQLite ledger records whether each article and figure has been retrieved, parsed and checked, so interrupted runs resume cleanly and every record can be traced to its source. Each case record holds the PMCID, figure number, image, caption, case history and reference diagnosis.

2.3 Preventing leakage

Captions often name the diagnosis outright (a caption of the form "Lesions of [diagnosis] on the trunk" is typical). In this corpus such captions are masked in the model's input but kept in the record, so they remain available for scoring and tracing. Table 1 lists the pitfalls and separates what was done during curation from what I recommend for the evaluation stage.

Table 1: Curation pitfalls and their handling.

Pitfall	Handling	Status
Diagnosis stated in caption	Mask in model input; keep in record for scoring and traceability.	Done
Mixed licenses	Record the license per article; use only content whose license permits the intended use.	Done
Interrupted crawls	Resume from the ledger; mark a record ready only after it passes the checks.	Done
Diagnosis in title or discussion	Expose only the image and the clinical history to the model.	Recommended
One patient, several figures	Keep all figures of a case on the same side of any split.	Recommended
Near-duplicate cases	Exclude the test article and its near duplicates from any retrieval pool.	Recommended

These choices matter for the proposal below: without traceable records there is nothing to cite, and without leakage control a grounding test measures nothing.

3 Proposal: DermTrace

3.1 Design

DermTrace splits the task into four roles that pass typed, versioned records to one another (Figure 1).

1. **Perception (VLM).** Describe before diagnosing. The VLM outputs structured findings in dermatology vocabulary (primary lesion, color, border, distribution, configuration), and each finding carries the image region it is drawn from. It is not asked for a diagnosis at this stage.
2. **Retrieval.** Find reference cases in the curated corpus that share the observed findings, using the structured findings together with text and image similarity. The test case's own article and its near duplicates are excluded.

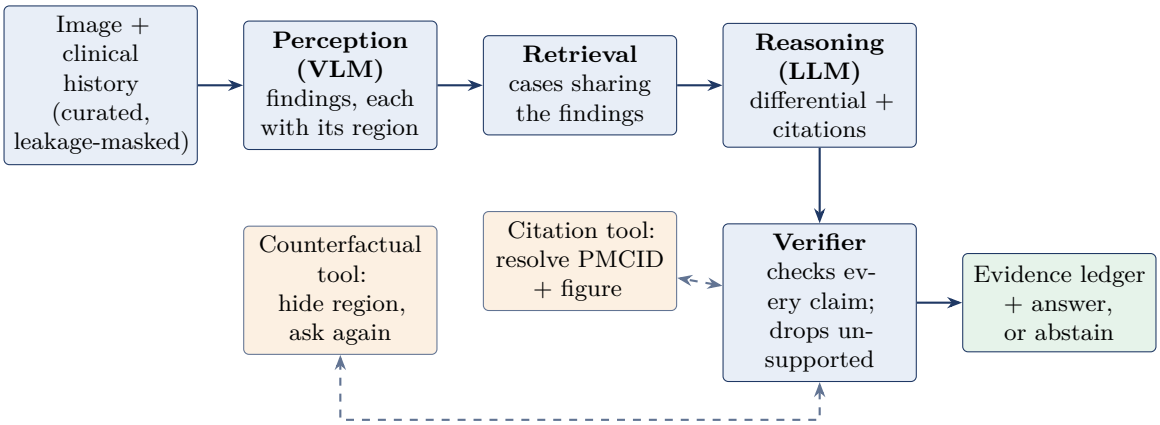


Figure 1: DermTrace. A VLM describes findings and the regions they come from; retrieval finds reference cases that share those findings; an LLM proposes a differential with citations; a verifier checks every claim with two tools before anything is shown. Unsupported claims are revised or dropped, and the system may abstain.

3. **Reasoning (LLM).** Build a ranked differential diagnosis. Each candidate must point to the findings that support it and cite reference cases, by PMCID and figure number, that share its diagnostic criteria.
4. **Verifier.** Check every claim before release, using two tools. If evidence is insufficient, return the evidence gathered so far and say what is missing, instead of a confident diagnosis.

3.2 Three ideas that make it different

(1) Counterfactual grounding, per finding. Existing grounding tests remove the whole image and compare accuracy [4]. DermTrace applies the test to each finding: the verifier hides the region a finding points to (filled from the surrounding skin) and asks the VLM again. A finding that survives the removal of its own evidence is not grounded and is dropped, together with any diagnosis that depended on it. To separate a real effect from the artifact of masking, the same test is run with an equal-sized region elsewhere on normal skin.

(2) Criteria-matched citations. A cited case is useful only if it shares the criteria that justify the diagnosis, not merely a similar look. The citation tool first checks that each cited PMCID and figure exist in the pool; the verifier then checks whether the cited case's documented findings overlap with the findings used in the argument. A natural-language inference model can screen this automatically, with a reviewer confirming a sample.

(3) Verify before release, and allow abstention. This borrows directly from two systems I have built: an enterprise RAG platform that only answers from retrieved evidence and links each answer back to its source, and GEO-Loop, a multi-agent pipeline that counts a step as done only after it is verified. In DermTrace the output is an *evidence ledger*: each finding with its region, each diagnosis with its supporting findings and citations, and each check that was run. Abstention is a first-class outcome rather than a failure.

3.3 Research questions

RQ1. Does describing findings first and verifying them counterfactually make diagnoses depend more on the image, without lowering accuracy?

RQ2. Do criteria-matched citations support the diagnosis more often than citations chosen by image or text similarity alone?

RQ3. Does verification-gated abstention give better accuracy on the cases the system does answer, at a fixed answer rate, than confidence thresholds?

4 Evaluation plan

Data. The curated PMC corpus, split by article so that no case appears on both sides; an external set such as DermCase [2] where its license allows.

Systems compared. (a) VLM answering directly; (b) VLM with step-by-step prompting; (c) VLM with text retrieval [10]; (d) a multi-agent baseline with roles but without the verifier, in the spirit of [7]; (e) DermTrace; and ablations of DermTrace without counterfactual checks, without criteria matching, and without abstention. An oracle condition gives the model hand-verified evidence, to separate retrieval failures from failures to use evidence.

Metrics.

- top-1 and top-3 diagnostic accuracy, and quality of the differential;
- visual reliance: the drop in accuracy when the lesion is hidden, compared with the drop when an equal-sized normal region is hidden;
- grounded-finding rate: the share of reported findings that pass the counterfactual test;
- citation validity (cited cases exist) and criteria support (cited cases share the criteria);
- selective accuracy: accuracy on answered cases at fixed answer rates, and the abstention rate.

A dermatologist's review of a sample of evidence ledgers would test whether the verifier's judgments match clinical judgment.

5 Risks and limitations

- **Masking artifacts.** Filling a region can create cues of its own; the control region is there to measure this, and the method may need a better inpainting step.
- **History still leaks.** Even with captions masked, the clinical history can point strongly to the diagnosis. Reporting results with and without history separates image evidence from text evidence.
- **Automatic criteria checks are imperfect.** Inference models make mistakes on clinical text; their output is a screen, not a verdict.
- **Case reports are unusual by design.** They over-represent rare presentations, so results should not be read as performance on everyday clinical photographs.
- **Cost.** Counterfactual checks multiply the number of model calls; the verifier should run them only on findings that drive the final diagnosis.

6 Status

The corpus work is done and was integrated into the team's pipeline. DermTrace is a proposal; I have not yet run it. The next step is a small pilot on held-out articles comparing systems (a), (d) and (e) on visual reliance and citation validity.

References

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